

11.17.16.3. Example: Hub-like Node

These types of Nodes are normally relatively capable, high-powered and always-on. These would most likely include time zone support for scheduling and display. Often these will have significant software beyond this specification and are likely to already have built-in mechanisms for time synchronization. Such Nodes SHOULD support an NTP server to serve time to other Nodes in the network. This type of Node may opt to support the TimeSyncClient (TSC) and NTPClient (NTPC) as backup, but can omit these if it has a reliable time source. It would support the following attributes based on the supported features:

- UTCTime
- Granularity
- TimeSource (optional)
- TrustedTimeSource (if TSC supported)
- DefaultNTP (if NTPC supported)
- TimeZoneListMaxSize
- TimeZone
- DSTOffsetListMaxSize
- DSTOffset
- LocalTime
- TimeZoneDatabase
- NTPServerAvailable

To achieve time synchronization, these Nodes would likely use Step 1 or Step 2 of [Time source prioritization](#), using the remaining options as fallback if necessary. The Node would then set its Granularity and TimeSource as appropriate, and would maintain its times, Granularity and TimeSource. The Node SHOULD also start an NTP server, update the NTPServerAvailable attribute and advertise using DNS-SD on _ntp._udp.

If a time-zone database is supported (Node can calculate DST times from TimeZone settings), these Nodes MAY subscribe to the DSTTableEmpty Events of Nodes with no TimeZoneDatabase support. Upon receipt of these events the Node SHOULD calculate and set new DST values for such Nodes by writing the DSTOffset attribute. These nodes MAY also subscribe to the TimeFailure and MissingTrustedTimeSource events for Nodes in its fabric and supply time or time sources as required.

11.18. Operational Credentials Cluster

This cluster is used to add or remove Node Operational credentials on a Commissionee or already-configured Node, as well as manage the associated Fabrics.

11.18.1. Revision History

The global ClusterRevision attribute value SHALL be the highest revision number in the table below.